Problem Statement and Goals
DAIIM

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Table 1: Revision History

Date	Developer(s)	Change
September 24	Robert	Write problem statement and inputs/outputs
September 25	Nathan	Write goals and stretch goals
September 27	Aidan	Added stakeholders and environment
September 27	Keira	Write custom documents and reflection
September 28	Robert	Revise problem statement
September 28	All	Review document and fix formatting
October 10	Robert	Ensure goals support claims made in prob statement. Ensure all goals are measurable.
October 26	Keira	Made Section 1.3 inputs/outputs less abstract
October 27	Robert	Fix typos and grammar inconsistencies.
November 4	Keira	Fix stakeholder claims not being in scope

1 Problem Statement

1.1 Problem

Current AI inference markets are increasingly concentrated around a small number of large organizations [1]. As the artificial intelligence (AI) industry continues to grow, we become increasingly reliant on centralized providers for AI models, data, hardware, and compute. This centralization increases the risk of biased outputs because of the lack of diverse data and training algorithms used by these companies [2]. Furthermore, the oligopolistic nature of the inference

market creates a risk of censorship and manipulation of model outputs. The lack of transparency from these companies and the black-box nature of many AI models make it difficult for consumers to verify the quality and reliability of the inferences they receive. Lastly, the centralized nature of the compute infrastructure used by these companies creates potential single points of failure. According to the 2026 Stanford AI Index, NVIDIA accounts for over 60% of total global AI compute capacity [3]. This concentration poses a significant risk of disruption to AI markets in the event of a single failure.

Our project is called DAIIM: Decentralized Artificial Intelligence Inference Marketplace. Our aim is to provide consumers with accurate AI predictions through a decentralized network of providers, reducing the reliance on centralized solutions.

1.2 Inputs and Outputs

Task requesters and inference providers are the primary users of DAIIM (see Section **??**).

The inputs to the system consist of task definitions, inference requests, network participation, and token stakes.

Task definitions are specifications of a problem to be solved via machine learning, including the expected input and output types for a machine learning model. Task definitions may be created by any user of the system and serve as a shared contract that defines what task requesters must provide and what inference providers must return.

An inference request is an instance of a task definition. Inference requests are submitted by task requesters and include all necessary data required for inference providers to run their models, in accordance with the associated task definition.

Network participation includes machine learning predictions for individual inference requests that are supplied to the network by individual inference providers.

Tokens are input into the system by requesters as payment for the request and providers participating in an inference request as collateral for the prediction accuracy.

The outputs of the system include an aggregated inference result, information about the network, and token payouts.

The system outputs an aggregated inference result related to an inference request to the original task requester, based on the predictions from participating providers. The aggregated inference result includes the final prediction for a given inference request, as well as information about the quality of the prediction.

Information about the network is also maintained in the system and output for transparent viewing for all users, including active providers and task definitions.

Tokens are output to users participating in an inference request in relation to the status of that inference request.

1.3 Stakeholders

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- **Task requesters**
  Task requesters are people who are requesting inferences to be made for a given task. These people value the decentralized nature of our platform which removes the risk of potential bias and a single point of failure. Some task requesters may value data privacy when making inference requests. For example, healthcare researchers would like to make requests to predict various regression inferences (such as biomarker levels) while still complying with privacy constraints. Similarly, decentralized finance platforms would like to make requests to predict various regression inferences (such as stock prices) while still complying with privacy constraints.

- **Inference providers**
  Inference providers are people with computing resources and models who want to monetize their regression models but lack the customer base, infrastructure, and visibility needed to compete with large established AI companies. These people usually have specialized or powerful models that can be used to make predictions for a given task.

- **Data providers**
  Data providers are people with data that can be used to train models for a given task. They are interested in helping the inference providers improve their models by providing them with data to train on. They want to be compensated and credited for their data, but lack a transparent way to share it with model developers and to track how it is used.

1.4 Environment

This software is compatible with any computer running a modern browser that is connected to the internet. There is no hardware component.

2 Goals

This project has five primary goals.

2.1 Transparency and Decentralization

Enable inference requesters and model providers to participate in a transparent inference marketplace that does not rely on a single centralized entity to manage its core operations. The marketplace's core logic, including payments, ownership, and settlement, will be implemented through smart contracts deployed on

a public blockchain. This provides publicly verifiable records of inference requests, provider submissions, aggregated results, provider reputation updates, and reward distributions. The system should support participation by multiple independent providers for each inference request, reducing reliance on a single model and allowing predictions from diverse sources to contribute to the aggregated result.

This goal will be evaluated through end-to-end testing. Success will require at least one inference request to proceed from submission through provider participation, result aggregation, and settlement. At least two independent providers must be able to participate in fulfilling a request. A tester must also be able to verify the relevant request, submission, aggregation, and settlement records using on-chain data. The tests must demonstrate that the smart contracts execute the required marketplace operations and settlement logic without requiring a centralized authority to approve each transaction.

2.2 Web3 Usability

Provide a user-friendly Web3 experience that hides unnecessary blockchain complexity while maintaining transparency into the actions and transactions performed by the system. The application should clearly communicate the purpose and consequences of actions requiring user authorization, including the transactions and data being signed, while providing users with direct visibility into the corresponding on-chain transactions, their status, and relevant state changes.

This goal will be evaluated through usability testing in which participants complete representative Web3 tasks, such as authorizing a transaction, checking its status, and locating the corresponding on-chain record. Participants will also answer questions about the purpose and consequences of the transactions they authorize. Success will be measured using task-completion rate, transaction understanding, and perceived ease of use. The target is for at least 80% of participants to complete the representative tasks without assistance, at least 80% to answer the transaction-understanding questions correctly, and the average ease-of-use rating to be at least 4 out of 5.

2.3 Reliable Inferences

Enable users to receive accurate and reliable results by aggregating redundant predictions from multiple model providers rather than relying on a single prediction. Provider reputation scores will reflect observed performance and influence the aggregation process, allowing contributions from providers with stronger records to have greater influence on the aggregated result. As these scores are updated over repeated requests, the network can adapt to provider performance and adjust the influence of their predictions. Having multiple registered providers should also improve the availability of inference services by allowing requests to be fulfilled when some providers are unavailable.

Reliability will be evaluated using the request-completion rate, defined as the percentage of submitted inference requests that receive an aggregated re-

sult within a predefined time limit. Tests will vary the number of available and participating providers and include scenarios in which some providers are unavailable. The completion rate will be measured under each scenario to assess how provider availability affects the system's ability to return results. Under the normal test configuration, the target is for at least 90% of submitted requests to receive an aggregated result within the specified time limit.

Prediction accuracy will be evaluated separately by comparing aggregated predictions against known ground-truth outcomes using a metric appropriate to the inference task. For example, a regression task may be evaluated using mean absolute error, while a classification task may be evaluated using accuracy or F1 score. The aggregated predictions will be compared with at least one baseline, such as predictions from an individual provider or an unweighted aggregation of provider predictions, to determine whether the implemented aggregation mechanism improves prediction quality under the tested conditions. The evaluation will also examine how provider reputation scores influence the aggregated results.

2.4 Accessible Inference Experience

Design an accessible user interface that enables users without machine-learning experience to submit the inputs required for an inference task and understand the resulting aggregated prediction and accompanying information. The interface should use clear signifiers for available actions, appropriate mappings between task inputs and controls, constraints that help prevent invalid submissions, and feedback that communicates request progress and results.

This goal will be evaluated through usability testing with participants unfamiliar with machine learning. Participants will complete representative tasks, including submitting an inference request, monitoring its progress, and interpreting the resulting aggregated prediction. Evaluation measures will include task-completion rate, time on task, user errors, and result comprehension. Success will require at least 80% of participants to complete the representative tasks without assistance and at least 80% to answer predefined questions about the aggregated results correctly. The evaluation will also identify errors that prevent successful task completion or lead to an incorrect interpretation of the result. The observed completion times and errors will be recorded to identify areas for improvement in the interface.

2.5 Model Provider Innovation

Encourage independent model providers to participate and improve their models within the marketplace by providing opportunities to make their models available to consumers, compete for inference requests, and receive compensation for their services. The marketplace should create incentives for providers to improve the quality, performance, availability, and cost-effectiveness of their models, as providers that deliver greater value to consumers may be able to attract more inference requests and earn greater rewards. Improvements in

provider performance may also improve the quality of the network's aggregated predictions.

This goal will be evaluated through controlled experiments using simulated providers with different prediction-quality levels and reputation scores. The experiments will assess whether the reward and reputation mechanisms respond as intended to differences in provider performance. In particular, under controlled conditions, the expected rewards of a provider that improves its prediction quality will be compared with the rewards it would receive without that improvement. The experiments will also track provider reputation scores and aggregated prediction quality across repeated inference rounds. Aggregated predictions will be evaluated against known ground-truth outcomes using a task-appropriate metric and compared with a baseline to determine whether improved provider predictions lead to improved aggregate prediction quality under the tested conditions. These experiments will establish whether the implemented incentive mechanisms reward improved provider performance and whether such improvements can benefit the network, without assuming that simulated improvements demonstrate real-world provider innovation.

3 Stretch Goals

This project has two stretch goals, which outline additional features that could be implemented after the primary goals have been achieved.

3.1 Expanded AI Model and Inference Support

Extend the marketplace beyond regression to support additional categories of AI inference and more complex tasks. This is considered a stretch goal because supporting multiple model architectures and more complex inference workflows would require additional development beyond the core marketplace functionality established by the primary goals.

3.2 Private AI Inference

Enable inference on private or sensitive data while protecting that data from unauthorized access during execution through the use of Trusted Execution Environments (TEEs). This is considered a stretch goal because incorporating TEEs would introduce additional security, infrastructure, and integration requirements beyond those necessary to support the core marketplace and inference functionality.

3.3 Data Provider Participation

Enable data providers to contribute training data to the marketplace and receive compensation when that data is used by inference providers to train or improve their models. Data contributions and their use would be recorded on-chain,

allowing data providers to be credited and rewarded in a transparent manner. This is considered a stretch goal because it would require mechanisms for data submission, storage, access control, and attribution of model improvements to specific datasets, all of which extend beyond the core inference marketplace established by the primary goals.

4 Custom Documents (CDs)

4.1 Cognitive Walkthrough

A cognitive walkthrough will be performed using a wireframe prototype of the planned interface of the application from the point of view of the inference requesters. This will aid in mapping the entire process of DAIIM and generally smooth out any remaining ambiguity. The results of the cognitive walkthrough will provide essential insight into the required user feedback and how to make the UI accessible and learnable for the expected users to ensure the application is simple to understand and enjoyable to use.

4.2 Business Plan

A business plan will be developed once a proof of concept has met expectations. The goal of the business plan is to detail how the decentralized AI marketplace can be implemented such that developers and stakeholders can be compensated, and will further explain the business strategy behind the tokenomics. The business plan would include a description of the problem and the market need, the market size and business model, the cost breakdown, the competition and positioning, and the execution plan and milestones.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. **What went well while writing this deliverable?**

   Our team was incredibly agreeable and had no trouble communicating our ideas for the project or our concerns about the scope. The team worked smoothly to accomplish the requirements for the deliverable. Everyone cooperated with one another, and it was easy to split up the workload evenly without disagreements or conflict. We were able to easily schedule multiple group meetings, which facilitated group discussions that gave everyone a deeper understanding of the project.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   There were varying levels of experience in the subject matter of ML, blockchains, inferences, and decentralized AI systems. Members who were less familiar with the content did some research on the subject matter, and members who were more experienced were extremely cooperative in answering questions in a non-judgmental and non-condescending manner. Notes of previous research performed by Robert were also shared as supplementary resources. Our team organized a couple of meetings to discuss the content in this document to ensure everyone was comfortable with the material and to encourage necessary discussion to determine the problem, users, goals, and scope of the project.

3. **How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?**

   Our team discussed our skills and experience, and ideas for features of the application. We considered the timeline and budget, and ultimately we think we can reasonably accomplish the project over the next few terms given our skill and experience levels.

Initially, we brainstormed goals that we later deemed to be too ambitious for the scope of the project. For example, we had planned to try to implement multiple types of inferences. Upon further inspection, we realized that not only was this goal too solution-oriented, but it was also beyond what we could determine was reasonably feasible within the timeline of the project, considering the other features we are planning to implement. We instead included this as a stretch goal.

References

[1] Jai Vipra and Anton Korinek. Market Concentration Implications of Foundation Models. https://arxiv.org/abs/2311.01550, 2023.

[2] Esther Rolf, Theodora T. Worledge, Benjamin Recht, and Michael Jordan. Representation Matters: Assessing the Importance of Subgroup Allocations in Training Data. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 9040–9051, 2021. https://proceedings.mlr.press/v139/rolf21a.html.

[3] Stanford Institute for Human-Centered Artificial Intelligence. The 2026 AI Index Report. Technical report, Stanford University, 2026. https://hai.stanford.edu/ai-index/2026-ai-index-report.